

WFH Burnout Predictor

- Machine Learning and Data Mining project following CRISP-DM.
- Goal: estimate burnout in work-from-home settings from daily behavioral indicators.
- Main task: regression on burnout_score.
- Secondary task: classification of burnout_risk.

Simone Mastria - University of Bologna

Assignment Fit

- Public dataset from Kaggle.
- Preprocessing is required: feature engineering, encoding, scaling and grouped split.
- Notebook runs end-to-end from the repository root and is Colab-ready.
- Deliverables: notebook, IEEE-style report, presentation and GitHub repository.

Dataset

Property	Value
Rows	1,800
Original columns	11
Users	180
Records per user	10
Missing values	0
Regression target	burnout_score
Classification target	burnout_risk

Data Understanding

- burnout_risk is highly imbalanced: Low 1527, Medium 253, High 20.
- task_completion_rate has correlation around -0.96 with burnout_score.
- No duplicates or missing values were found.
- Outliers are detected but not automatically removed because extreme burnout cases may be meaningful.

Data Preparation

- user_id is excluded from predictive features.
- user_id is used as group variable for train/test split and cross-validation.
- Numeric features are scaled; categorical features are one-hot encoded.
- Feature engineering adds workload intensity, meeting density, sleep deficit and overwork flag.
- All transformations are inside scikit-learn pipelines to avoid leakage.

Modeling Strategy

- Baselines: DummyRegressor and DummyClassifier.
- Regression models: Ridge, KNN, Random Forest, Hist Gradient Boosting.
- Classification models: balanced Logistic Regression, Random Forest and Hist Gradient Boosting.
- Hyperparameter tuning is limited and uses grouped cross-validation.

Regression Results

Model	Test MAE	Test RMSE	Test R2
Ridge	4.79	6.31	0.934
Random Forest	4.85	6.41	0.932
Hist Gradient Boosting	5.14	6.73	0.925
KNN	9.26	11.65	0.776
Dummy median	20.35	25.80	-0.099
Tuned Random Forest	4.88	6.21	0.936

Sensitivity Analysis

- task_completion_rate dominates the prediction.
- Removing it increases MAE to about 20 for Ridge and Random Forest.
- This means the model is predictive, but not evidence of causal burnout mechanisms.
- This is the most important limitation to explain during the oral exam.

Classification and Fairness

- Best classifier: balanced Logistic Regression.
- Balanced accuracy: 0.803; macro F1: 0.646.
- High-risk support in test set is only 4, so conclusions are fragile.
- Fairness checks compare errors across operational groups such as `day_type`, `after_hours_work` and work-hour bands.

Conclusions

- The project is a complete CRISP-DM workflow, not only model training.
- Regression is the most reliable task formulation.
- Classification is useful for interpretation but limited by imbalance.
- Real deployment would require external validation, fairness governance and human oversight.
- The model must not be used for automatic HR decisions.